Split curvature and convergence of B-adaPG with flat boundary kernels

Abstract

We prove convergence of the original undamped adaptive Bregman proximal-gradient method under additional curvature in the finite proximal term. The kernel may have a flat boundary face through the optimizer. The proximal term is only required to be strictly convex on the feasible set, with no strong-convexity modulus; the differentiable term may be nonlinear, but its relative smoothness must be uniform on bounded interior sets approaching the boundary. Retaining a nonsmooth slack in the exact merit identity gives separation through compact midpoint gaps. Divergence of cumulative selected stepsizes and a small-step/large-step argument then imply objective and point convergence for arbitrary positive initialization. A nonlinear example verifies the assumptions with a flat kernel face. The unrestricted unique-minimizer boundary problem remains unresolved.

Keywords: adaptive Bregman proximal gradient; flat boundary kernels; split curvature; relative smoothness; nonsmooth convex penalties; point convergence

2020 Mathematics Subject Classification. Primary 65K10; Secondary 90C25, 49J52.

1 Setting and exact controller

For a differentiable convex function v, write $D_v(a,b) = v(a) - v(b) - \langle \nabla v(b), a-b\rangle$, with the second argument in its differentiability domain. The conjugate is $h^*(p) = \sup_x\{\langle p, x\rangle - h(x)\}$. We use the Euclidean norm. A function is 1-coercive if $h(x)/\|x\| \to +\infty$ as $\|x\| \to \infty$. Legendre means essentially smooth and essentially strictly convex.

We study the original undamped B-adaPG controller of Ou, Latafat and Themelis [1], pinned to arXiv:2508.01353v2. Our added assumptions are stated explicitly below. The proof retains nonsmooth curvature rather than assuming strict boundary geometry of the kernel.

1.1 Hypotheses

Let $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous, convex, 1-coercive and Legendre. Let $C = \operatorname{int}\operatorname{dom} h \neq \varnothing$ and $K = \operatorname{dom} h = \overline{C}$. Assume $h \in C^2(C)$ with positive definite Hessian, and the following boundary properties (Assumption 2.4 of the cited paper):

1. $D_h(z, y_i) \to 0$ whenever $z \in K$, $y_i \in C$, and $y_i \to z$.
2. For every $z \in K \setminus C$, the map $y \mapsto D_h(z, y)$ is level bounded.

No strict convexity of h on K or on the feasible closure is assumed.

Let $U \supset K$ be open and convex and $f : U \to \mathbb{R}$ be convex and continuously differentiable. Require the following stronger-than-source relative smoothness:

For every $R > 0$ there exists $L_R > 0$ such that $L_R h - f$ is convex on $C \cap B(0, R)$.

Thus the modulus is uniform up to boundary approaches inside a fixed bounded ball. Let $g_0 : \mathbb{R}^n \to \mathbb{R}$ be finite convex, and D be closed convex with $D \cap C \neq \varnothing$. Write $S = K \cap D$ and $F = f + g_0$ on S. Suppose F attains its minimum and $g_0|_S$ is strictly convex. This implies a unique minimizer z. No strong-convexity modulus is required.

1.2 Exact initialized algorithm

Use $g = g_0 + I_D$ and the original ambient, undamped B-adaPG, equations (2.3) and (2.6) of Ou–Latafat–Themelis, arXiv:2508.01353v2. Start at arbitrary $x_0 \in C \cap D$ and arbitrary $\gamma_0, \gamma_1 > 0$. The first update uses γ_1, and every update is

$$x_{k+1} = \operatorname*{argmin}_{y \in S} \left\{ g_0(y) + \langle \nabla f(x_k), y - x_k \rangle + \frac{D_h(y, x_k)}{\gamma_{k+1}} \right\}.$$

For $k \geq 1$ and distinct consecutive iterates define

$$\rho_k = \gamma_k/\gamma_{k-1}, \quad r_{k+1} = \sqrt{1 + \rho_k}, \quad \delta_{k+1} = 2r_{k+1}, \quad H_k = \nabla h - \gamma_k \nabla f,$$
$$\ell_k = \frac{\Delta_f(x_k, x_{k-1})}{\Delta_h(x_k, x_{k-1})}, \qquad \alpha_k = \frac{D_h(x_k, x_{k-1})}{D_h(x_{k-1}, x_k)},$$
$$\Lambda_k = \frac{2D_{h^*}\big(\nabla h(x_k) + \delta_{k+1}[H_k(x_k) - H_k(x_{k-1})], \nabla h(x_k)\big)}{\delta_{k+1}^2 \Delta_h(x_k, x_{k-1})},$$
$$\rho_{k+1} = \min\left\{ r_{k+1}, \frac{\alpha_k}{(1 + \alpha_k) 2r_{k+1}[\Lambda_k - (1 - \gamma_k \ell_k)]_+} \right\}, \qquad \gamma_{k+1} = \rho_{k+1}\gamma_k.$$

Here $\Delta_v(a, b) = D_v(a, b) + D_v(b, a)$, $[t]_+ = \max(t, 0)$ and positive divided by zero is $+\infty$. The conjugate h^* and all controller vectors are full ambient quantities; neither projection nor kernel restriction is allowed. If consecutive iterates coincide, stop before evaluating zero denominators and extend the sequence constantly.

2 Convergence theorem

Theorem 1. *Under the hypotheses of Section 1, every proximal update is uniquely defined in $C \cap D$, and every nonstopped generated stepsize is finite and positive. The exact orbit satisfies*

$$F(x_k) \longrightarrow F(z), \qquad x_k \longrightarrow z.$$

No stepsize floor, upper bound, damping, boundary avoidance, bounded constraint normals, affine-f assumption, or strict boundary-kernel geometry is imposed. No rate is asserted.

If an ambient extended-real objective is desired, extend f outside U by the supremum of its affine supports from U. This changes no feasible value or algorithm gradient.

3 Well-definedness and stopping

The support extension of f is proper closed convex and agrees with f throughout U. Every proximal problem is equivalent to minimizing

$$h(y) + \gamma g_0(y) + I_D(y) + \langle \gamma \nabla f(x) - \nabla h(x), y \rangle.$$

An affine lower support of finite convex g_0 and 1-coercivity of h give coercivity. Lower semicontinuity and feasibility give a minimizer. Because $C \cap D \neq \varnothing$ and C is open, $C \cap \operatorname{ri} D \neq \varnothing$. The exact convex sum rule and essential smoothness of h exclude any minimizer outside C, since $\operatorname{dom} \partial h = C$. Strict convexity of h on C then gives uniqueness. At each update there exist $s_{k+1} \in \partial g_0(x_{k+1})$ and $n_{k+1} \in N_D(x_{k+1})$ with

$$p_{k+1} - p_k = -\gamma_{k+1}(\nabla f(x_k) + s_{k+1} + n_{k+1}), \qquad p_k = \nabla h(x_k). \tag{1}$$

The full conjugate of a 1-coercive Legendre function is finite and continuously differentiable on all of $\mathbb{R}^n$. Thus every ambient controller probe is valid. For distinct interior endpoints both kernel divergences are strictly positive. Every nonstopped controller value is finite where required; its first ratio branch is positive finite and the second positive or infinite. Induction proves positive finite generated stepsizes. A repeated iterate in (1) satisfies the constrained convex optimality condition, hence is optimal on S. Strict convexity of $g_0|_S$ makes the minimizer unique. We henceforth consider an infinite nonstopped run.

4 Retaining the nonsmooth slack

For $j \geq 1$ select the actual full subgradient $\xi_j = s_j + n_j \in \partial(g_0 + I_D)(x_j)$ supplied by (1), and write

$$\widetilde{D}_g(u, x_j) = g(u) - g(x_j) - \langle \xi_j, u - x_j \rangle \geq 0.$$

Set $m = F(z)$, $P_k = F(x_k) - m \geq 0$. For $k \geq 2$ define

$$d_k = 1 - \frac{\rho_k}{2r_k}, \qquad V_k = D_h(z, x_k) + \gamma_k(1 + r_k)P_{k-1} + d_k D_h(x_k, x_{k-1}).$$

Define also $\widetilde{D}_F(u, x_k) = F(u) - F(x_k) - \langle \nabla f(x_k) + \xi_k, u - x_k \rangle$. It is nonnegative by convexity. To give the full algebra behind the retained slack, put $\vartheta = r_{k+1}$ and

$T_k = (1 - \gamma_k \ell_k)\Delta_h(x_k, x_{k-1})$. The optimality equation and the Bregman three-point identity give

$$D_h(z, x_{k+1}) + \gamma_{k+1} P_k + D_h(x_{k+1}, x_k) = D_h(z, x_k) + B_{k+1} - \gamma_{k+1} Q_k,$$
$$P_{k-1} - P_k = T_k/\gamma_k + \widetilde{D}_F(x_{k-1}, x_k),$$

where $Q_k = D_f(z, x_k) + \widetilde{D}_g(x_{k+1}, x_k) + \widetilde{D}_g(z, x_{k+1})$. For the second equality, use $\nabla f(x_k) + \xi_k = -[H_k(x_k) - H_k(x_{k-1})]/\gamma_k$ and take its inner product with $x_{k-1} - x_k$. Adding $\gamma_{k+1}\vartheta$ times the second equality to the first yields the exact identity

$$\begin{aligned} &D_h(z, x_{k+1}) + \gamma_{k+1}(1+\vartheta) P_k + D_h(x_{k+1}, x_k) \\ &\quad = D_h(z, x_k) + \gamma_{k+1}\vartheta P_{k-1} - \rho_{k+1}\vartheta T_k + B_{k+1} \\ &\qquad - \gamma_{k+1}\{Q_k + \vartheta \widetilde{D}_F(x_{k-1}, x_k)\}. \end{aligned}$$

This is Lemma 3.1 of the cited paper, derived here with the actual selected subgradients. Every term in the braces is nonnegative, even with the constraint indicator. The cross term is

$$B_{k+1} = \rho_{k+1}\langle H_k(x_k) - H_k(x_{k-1}), x_{k+1} - x_k \rangle.$$

Apply the Bregman Young inequality at x_k with vector $\delta_{k+1}[H_k(x_k) - H_k(x_{k-1})]$ (equivalently the source's bound (3.11)). It gives

$$B_{k+1} \le \frac{\rho_{k+1}}{\delta_{k+1}} D_h(x_{k+1}, x_k) + \frac{\delta_{k+1}\rho_{k+1}}{2} \Lambda_k \Delta_h(x_k, x_{k-1}). \tag{2}$$

The primal displacement is $x_{k+1} - x_k$, matching the positive dual displacement in Λ_k; no divergence orientation is reversed. Substitution of $\delta_{k+1} = 2r_{k+1}$ and direct collection yield

$$V_{k+1} \le V_k - A_k P_{k-1} - B_k D_h(x_k, x_{k-1}) - \gamma_{k+1} \widetilde{D}_g(z, x_{k+1}), \tag{3}$$

where

$$A_k = \gamma_k[1 + r_k - \rho_{k+1} r_{k+1}],$$
$$B_k = d_k - \rho_{k+1} r_{k+1} \frac{1+\alpha_k}{\alpha_k}[\Lambda_k - (1 - \gamma_k \ell_k)].$$

For $k \ge 2$ the first selector branch gives $\rho_k \le r_k$, so $d_k \ge 1/2$. Also $\rho_{k+1} r_{k+1} \le r_{k+1}^2 = 1 + \rho_k \le 1 + r_k$, hence $A_k \ge 0$. If the bracket in B_k is nonpositive then $B_k \ge d_k$; if positive the second selector branch bounds the subtracted quantity by $1/2$, giving $B_k \ge 0$. Therefore V_k is nonnegative nonincreasing and

$$D_h(z, x_k) \le V_2, \qquad \sum_{k \ge 2} \gamma_{k+1} \widetilde{D}_g(z, x_{k+1}) < \infty. \tag{4}$$

Reverse-divergence level boundedness at z gives a bounded orbit. For interior z, the support-ball estimate proved below with $a = z$ bounds $\nabla h(x_k)$, and continuity of ∇h^* gives boundedness of x_k. For boundary z, boundedness is exactly the assumed boundary sublevel property. No vanishing-divergence-to-primal-distance implication is used.

5 Strict nonsmooth curvature separates endpoints on the compact feasible set

Choose a closed ball containing z and the orbit, and let Q be its intersection with S. This is compact and convex. Finite convex g_0 is continuous. Its midpoint gap

$$J_g(u, v) = \frac{g_0(u) + g_0(v)}{2} - g_0((u + v)/2)$$

is continuous, nonnegative, and strictly positive for distinct $u, v \in Q$. The full subgradient inequality at $v = x_j$ and midpoint $(z + v)/2 \in S$ gives

$$\widetilde{D}_g(z, v) \geq 2J_g(z, v). \tag{5}$$

This includes the normal component without estimating its norm. For every $\varepsilon > 0$ for which the indicated set is nonempty,

$$c_\varepsilon = \min\{2J_g(z, v) : v \in Q,\ \|v - z\| \geq \varepsilon\} > 0. \tag{6}$$

No quadratic lower bound or uniform strong convexity is inferred. Combining (4)–(6) proves that for every fixed $\tau > 0$, along all sufficiently late endpoint indices with $\gamma_j \geq \tau$ one has $x_j \to z$, and therefore $F(x_j) \to m$.

6 Divergence of cumulative stepsizes

We give the needed independent selector argument. Suppose $\sum_{j\geq 1} \gamma_j < \infty$. Fix $a \in C \cap D$. The three-point identity and (1), with the favorable normal sign, give for $x = x_k$, $y = x_{k+1}$

$$D_h(a, y) - D_h(a, x) \leq -D_h(y, x) + \gamma_{k+1}\{\langle \nabla f(x), a - y\rangle + g_0(a) - g_0(y)\}. \tag{7}$$

Boundedness of the orbit, continuity of g_0 and boundedness of ∇f on the relevant compact set make the braces uniformly bounded. Thus $D_h(a, x_k)$ is bounded. Choose $r > 0$ with $\overline{B}(a, r) \subset C$ and put $M = \max_{\overline{B}(a,r)} h$. The support inequality at x evaluated at $a + rp/\|p\|$, $p = \nabla h(x) \neq 0$, gives

$$D_h(a, x) \geq r\|p\| + h(a) - M.$$

Consequently p_k is bounded. Continuity of ∇h^* on the whole ambient dual space puts all $x_k = \nabla h^*(p_k)$ in a compact subset of C. A slightly enlarged compact convex neighborhood $E \subset C$ has uniform Hessian bounds $0 < \mu I \preceq \nabla^2 h \preceq M_h I$.

Proximal comparison with x_k, local Lipschitz continuity of g_0, and boundedness of ∇f yield

$$\frac{\mu}{2}\|x_{k+1} - x_k\|^2 \le D_h(x_{k+1}, x_k) \le B\gamma_{k+1}\|x_{k+1} - x_k\|.$$

Thus consecutive differences tend to zero. Relative smoothness on E and the Hessian upper bound imply a local Lipschitz bound for ∇f: indeed $(LM_h/2)\|x\|^2 - f$ is convex on E, and the usual convex smoothness equivalence applies on its neighborhood.

Let $t_k = x_k - x_{k-1}$ and $A_k^h = \nabla^2 h(x_k)$. Uniform Taylor expansion on E gives

$$p_k - p_{k-1} = A_k^h t_k + o(\|t_k\|), \quad \Delta_h(x_k, x_{k-1}) = t_k^T A_k^h t_k + o(\|t_k\|^2), \quad \alpha_k \to 1.$$

Put $R = \max\{\rho_1, (1+\sqrt{5})/2\}$. Induction using the first selector branch gives $\rho_k \le R$, $1 \le r_{k+1} \le R$, hence $2 \le \delta_{k+1} \le 2R$. Since $\gamma_k \to 0$, bounded relative smoothness gives $\gamma_k \ell_k \to 0$, and Lipschitz continuity of the gradient gives

$$v_k := H_k(x_k) - H_k(x_{k-1}) = A_k^h t_k + o(\|t_k\|).$$

On a compact neighborhood of the bounded dual sequence, h^* is C^2 with Hessian at p_k equal to $(A_k^h)^{-1}$. Its uniform Taylor expansion gives

$$2D_{h^*}(p_k + \delta_{k+1} v_k, p_k) = \delta_{k+1}^2 v_k^T (A_k^h)^{-1} v_k + o(\|t_k\|^2).$$

Therefore $\Lambda_k \to 1$. The positive-part bracket in the second selector branch tends to zero while $\alpha_k \to 1$ and r_{k+1} is bounded. That ratio branch tends to $+\infty$. Eventually the first branch is selected, and $\gamma_{k+1} = r_{k+1}\gamma_k \ge \gamma_k > 0$, contradicting summability. Hence

$$\sum_{j \ge 1} \gamma_j = \infty. \tag{8}$$

This does not assert a global stepsize floor.

7 Objective convergence

Equations (4)–(6) and (8) imply $\liminf_j \|x_j - z\| = 0$: otherwise all sufficiently late endpoint slacks have a common positive lower bound, contradicting weighted summability. Thus there is a subsequence $x_j \to z$, and continuity gives $\liminf_j F(x_j) = m$.

Choose a ball strictly containing the bounded orbit and the associated fixed $L > 0$ from the uniform relative-smoothness hypothesis. For every pair of consecutive interior endpoints,

$$D_f(x_{k+1}, x_k) \le L D_h(x_{k+1}, x_k).$$

The proximal comparison with x_k gives

$$F(x_{k+1}) - F(x_k) \le (L - 1/\gamma_{k+1}) D_h(x_{k+1}, x_k). \tag{9}$$

Consequently every update with $\gamma_{k+1} \le 1/L$ is nonincreasing in objective. Every sufficiently late endpoint with $\gamma_{k+1} > 1/L$ has objective arbitrarily close to m, by Section 5 with $\tau = 1/L$.

Fix $\eta > 0$. Choose N so all those large-step endpoints at indices at least N satisfy $F(x_j) < m + \eta$. Choose a subsequence index $p \ge N$ with $F(x_p) < m + \eta$. Inductively, a small-step update preserves this bound by (9), and a large-step update satisfies it by the choice of N. Thus $F(x_j) < m + \eta$ for every $j \ge p$. As $F \ge m$ and η is arbitrary, $F(x_j) \to m$.

Every cluster lies in closed S and is optimal by continuity. The optimizer is unique, so boundedness implies $x_j \to z$. This proves both conclusions, without interpreting $D_h(z, x_j) \to 0$ as Euclidean convergence.

8 A nonlinear flat-kernel example

Put $e = 1/256$, $a = 1/100$, $K = [-1, 1] \times [0, e]$, and $C = \operatorname{int} K$. With $0 \log 0 = 0$, define

$$\begin{aligned} b(x) &= (1+x)\log(1+x) + (1-x)\log(1-x), \\ q(y) &= -\sqrt{y} - \sqrt{e-y}, \\ h(x,y) &= y[b(x) - ax\sin(\log y)] + q(y) \quad (y > 0), \\ h(x,0) &= -\sqrt{e}, \end{aligned}$$

and set $h = +\infty$ off K. For arbitrary $\beta, \kappa > 0$, take $f(x, y) = y + \beta y^2$, $g_0(x, y) = \kappa(x^2 + y^2)$ and $D = \mathbb{R}^2$. The kernel is constant on the entire bottom edge, while the forward objective is nonlinear. Continuity of b on $[-1, 1]$ and $y \sin \log y \to 0$ make h continuous on K. Its infinity extension is proper lower semicontinuous, and its bounded domain gives 1-coercivity. Write $t = \log y$, $B = b''(x) = 2/(1 - x^2)$ and $b'(x) = \log((1 + x)/(1 - x))$. On C,

$$h_{xx} = yB, \quad h_{xy} = b'(x) - a(\sin t + \cos t),$$
$$h_{yy} = q''(y) - ax(\cos t - \sin t)/y, \quad q''(y) = \tfrac{1}{4}[y^{-3/2} + (e - y)^{-3/2}].$$

Putting $u = \operatorname{atanh} x$, the inequality $\cosh u \ge \sqrt{2}|u|$ gives $b'(x)^2/B \le 1$. Since $B \ge 2$,

$$h_{xy}^2/B \le 2 + 2a^2, \qquad y h_{yy} \ge 4 - 2a.$$

Let $c_0 = 2 - 2a - 2a^2 > 0$. The Schur complement satisfies

$$h_{yy} - h_{xy}^2/h_{xx} \geq c_0/y \geq c_0/e > 0. \tag{10}$$

Thus the Hessian is positive definite. Continuity extends convexity to boundary chords. The derivatives

$$h_x = y[b'(x) - a\sin t], \quad h_y = b(x) - ax(\sin t + \cos t) + q'(y)$$

have divergent norm at every approach to the boundary: horizontal-edge divergence comes from q', with bounded remaining terms, and vertical-edge divergence away from horizontal edges comes from b'. Inward one-sided difference quotients are $-\infty$ on every edge, also at corners. Hence there are no boundary subgradients, and strict interior convexity establishes Legendreness.

For fixed $(X, Y) \in K$ and $(u, v) \to (X, Y)$ from C, continuity reduces Assumption 2.4(i) to

$$h_x(u, v)(X - u) \to 0, \qquad h_y(u, v)(Y - v) \to 0.$$

The first follows from $(X - u)b'(u) \to 0$ at vertical endpoints and bounded v. The second follows from bounded nonsingular terms and the square-root limits $(Y - v)q'(v) \to 0$ at $Y = 0, e$. These cover mixed corners. Reverse sublevels are bounded because K is bounded, proving Assumption 2.4(ii).

For $f = y + \beta y^2$, the only nonzero Hessian entry is $f_{yy} = 2\beta$. Equation (10) implies $L\nabla^2 h - \nabla^2 f \succeq 0$ throughout C whenever $L \geq 2\beta e/c_0$. Thus the stronger uniform relative smoothness holds globally, not merely along an assumed orbit. The finite quadratic $g_0 = \kappa(x^2 + y^2)$ is strictly convex. On K, $F = y + (\beta + \kappa)y^2 + \kappa x^2 \geq 0$, with equality only at $(0, 0)$. All hypotheses therefore hold. Since every finite iterate has positive y, no finite iterate is optimal or invokes stopping.

9 What is and is not established

This is a nonlinear-forward-split theorem with a genuine flat kernel face through the unique optimizer. It permits nonlinear f and kernels that are not strictly convex on the feasible closure. The added requirements are strict convexity of the finite nonsmooth term on S and relative smoothness uniform on bounded interior sets approaching the boundary. These are substantially stronger than mere uniqueness of the total minimizer and the source's compact-interior relative smoothness. The general unique-minimizer boundary question remains unresolved.

References

[1] H. Ou, P. Latafat and A. Themelis. Linesearch-free adaptive Bregman proximal gradient for convex minimization under local relative smoothness. *arXiv:2508.01353v2*, 5 June 2026. `https://arxiv.org/abs/2508.01353v2`.